Planck-Scale Integration into Hologram Moonshine (Φ-Atlas)

Comprehensive Mathematical Note (APEX v1)

Scope. This note formalizes the Planck-scale modeling layer (UOR@IFMD) and its certified interface to the Π-kernel and the Multiplicity Runtime, culminating in acceptance through the Six-Level Tetrahedral Rhythm (R96, C768, Φ-compatibility). The stance is rigor-first and architectural (non-physical claims avoided). All quantities are unit-consistent; all budgets and certificates are explicit.

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0) Executive Summary

- **Discrete stance (Axiom A9):** Work "as if" spacetime is the 4D lattice $L = \{(n_1\ell_p, n_2\ell_p, n_3\ell_p, n_0ct_p) : n_i \in \mathbb{Z}\}$ with anisotropic sup metric. Queries are **processable** only if their kernels obey $\delta_{\text{spatial}} \geq \ell_p$ and $\delta_{\text{temporal}} \geq t_p$.
- **Certified control:** Execute at Planck resonance via **certificates** (SlopeUB, GapLB) within an ACE weighted-ℓ_1 budget; keep $\|K\|_2 \leq 1 - \varepsilon$ so $\rho(K) < 1$.
- **Telemetry & audits:** Model residuals by a causal-echo SPDE; audit on threshold crossings; close the loop via ACE projection and budget tightening.
- **Adapter to runtime:** Expose kernel steps as $T_{t+1} = F_t + K_tT_t$ with channelization, ACE projection, and PETC logging; use WAC/DSE envelopes to guarantee contraction on windows.
- **Moonshine boundary gates:** Prove the order-2048 boundary subgroup action, implement the canonical fair schedule C768 with closed-form sums, verify Φ-equivariance, and deterministically obtain exactly 96 resonance classes.

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1) Constants & Units

Let ℓ_p, t_p, E_p, c denote Planck length, time, energy, and the speed of light. Spatial extents use meters; time uses seconds; time-to-length conversions use ct when needed. All equations carry explicit units.

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2) Certified Lattice Model (UOR@IFMD)

2.1 Axiom A9 (Discrete Model Hypothesis)

We **adopt** A9 as a modeling stance: the system operates as if reality were discretized on the 4D lattice L above. This stance is purely methodological and does not assert ontology.

2.2 Metric (anisotropic sup)

For $x = (n_1\ell_p, n_2\ell_p, n_3\ell_p, n_0ct_p)$ and $y = (m_1\ell_p, m_2\ell_p, m_3\ell_p, m_0ct_p)$, define

$$d(x, y) = \max\{\max_{i=1..3} \ell_p|n_i - m_i|,\ ct_p|n_0 - m_0|\}.$$

This yields clean computability and unit consistency.

2.3 Processability Guard (boundary policy)

A query with measurement kernel Q_{K_δ} is **processable** iff

$$\delta_{\text{spatial}} \geq \ell_p, \quad \delta_{\text{temporal}} \geq t_p.$$

The guard is enforced at the boundary and composes with runtime guards.

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3) Certified Randomness & Quantization

- **QuantumFoam():** Provide near-uniform bits with a **certified min-entropy per block** via conservative estimation and a seeded extractor. Device-independent options may be used when available.
- **QCheck (Physical extent validation):** For each object/component, discretize extents to the lattice shells $L_j(D) = \{n_j\ell_p \pm \varepsilon : n_j \in \mathbb{Z},\ \varepsilon < \frac{\ell_p}{2}\}$. **QCheck(D)** returns true iff all extents satisfy tolerance (optionally mirrored to energies via E_p).
- **Information interface:** Quantify quantization loss $\Delta I = H(C) - H(Q(C))$ or use a Lipschitz-distortion bound on concept spaces.

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4) Planck Resonance via Certificates & ACE Budgets

We govern execution by **certificates** and a budget, not by energy heuristics.

4.1 SlopeUB and GapLB

For a generic step $x_{t+1} = f_t(x_t, u_t)$, define

$$\text{SlopeUB} := \sup_x \|\partial f_t/\partial x\|, \qquad \text{GapLB} := 1 - \rho(K_t),$$

where in an affine/linearized form $x_{t+1} = F_t + K_t x_t$. In LTI, SlopeUB$= \|K\|$ (chosen operator norm) and GapLB$= 1 - \rho(K)$. We **require** $\rho(K_t) < 1$, hence **positive** GapLB.

4.2 PIRTM ACE budget

Maintain a weighted-ℓ_1 inequality on channel weights $w = (w_p)_p$:

$$\sum_p b_p |w_p| \ \le \ \tau \ < 1.$$

This is enforced by projection (ACE) and adapted during audits.

4.3 Norm/radius relation

We use $\rho(K) \le \|K\|_2$ and the conservative bound $\mathrm{GapLB} \ge 1 - \|K\|_2$. When a sharper $\rho(K)$ estimator is available (e.g., Arnoldi), substitute it.

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5) Causal Echoes (Residual Modeling)

Let $e(x,t)$ be residual energy-density deviation [J/m³]. Generation variance obeys $\mathrm{Var}[e] \equiv \sigma_V^2 k_t$. Propagation is modeled by the SPDE

$$\partial_t e = D\,\nabla^2 e + \eta(x,t),$$

with η spatially white and temporally Ornstein–Uhlenbeck. Specify boundary conditions (e.g., Neumann) and calibrate D from telemetry.

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6) Audits & Responses (Closed-Loop Safety)

Triggers. Audit when SlopeUB crosses its threshold, GapLB falls below its threshold, or residuals exceed SPDE forecasts.

Responses. Tighten budgets ($\tau \mapsto \alpha\tau,\ 0<\alpha<1$), re-project via ACE, and re-estimate certificates until margins are restored.

Telemetry. Log SlopeUB, GapLB, τ , projection magnitudes, echo residuals (empirical vs. SPDE), and quantization histograms.

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7) Tiny Monitoring Stub (reference implementation)

A minimal monitor enforces the ACE budget, estimates SlopeUB/GapLB via $\|K\|_2$, and auto-tightens τ on audit. Replace the weighted-ℓ_1 projector with an exact routine if desired; the prototype enforces feasibility and works well in practice.

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8) Π-Kernel ↔ Multiplicity Runtime Adapter

8.1 Setting (atoms, frames, channels)

Let E be a real/complex Banach space. A family of idempotents $(R_\pi)_\pi$ indexes **Π-atoms** (Π-IDs). In the orthonormal case, $R_\pi R_{\pi'} = \delta_{\pi\pi'} R_\pi$ and $\sum_\pi R_\pi = I$. For tight frames allow an **orthogonality defect** $\delta \geq 0$ with standard frame bounds.

Let $\mathcal{P}$ be **prime channels**; a routing $r : \Pi \to \mathcal{P}$ induces bounded channel projectors $P_p = \sum_{\pi \in r^{-1}(p)} R_\pi$.

8.2 Adapter form and safety

Expose the kernel as a global affine update

$$T_{t+1} = F_t + K_t T_t, \qquad K_t = \sum_{p \in \mathcal{P}} w_{p,t} B_{p,t},$$

then **group by channels** and project w_t into the ACE safety set $\sum_p b_p |w_{p,t}| \leq 1 - \varepsilon_t$. When ACE keeps $\|K_t\| \leq 1 - \varepsilon_t$, the step is a **contraction** (with WAC/DSE envelopes below) and admits explicit rollback conditions.

8.3 Runtime invariants & PETC

The runtime logs per-step tuples (ΠID witness, invariant digests, SlopeUB, GapLB) to the PETC ledger and enforces - **Contraction margin** $\gamma_t := 1 - \sum_p b_p |w_{p,t}| \geq \varepsilon_t$, - **Projection KKT residual** small (feasible projection), - **WAC/DSE** window checks for long-horizon contraction, - **Lawfulness:** additivity and channel action of prime signatures; fail-closed on violation.

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9) Envelope Theorems (Runtime)

- **WAC (Windowed Average Contraction):** If products of K_t over windows of size m contract with rate $\rho \in (0, 1)$, then T_t converges at least geometrically at rate $\rho^{1/m}$.
- **DSE (Drift with Sparse Expansions):** Occasional $\|K_t\| > 1$ is permitted provided each macro-window contracts; conclusions as under WAC with adjusted parameters. These envelopes are compatible with ACE projection and the adapter form above.

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10) Category-Theoretic Framing (APEX view)

Model Π-atoms as objects in a rigid monoidal category with orthogonal direct sums preserved by a strong monoidal functor to prime channels. The runtime is a **contractive endofunctor** driven by the adapter; PETC is a natural transformation that preserves prime signatures. *(Exploratory gloss consistent with the formal adapter.)*

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11) Six-Level Tetrahedral Rhythm (Boundary Gates)

11.1 Boundary subgroup & tiling

Let pages $P=\{0,\ldots,47\}$, bytes $B=\{0,\ldots,255\}$, and $G=P\times B$ ($|G|=12{,}288$). Define commuting involutions by XOR on low page bits and all byte bits. The **canonical subgroup** $U_{\mathrm{ref}}\cong(\mathbb{Z}/2\mathbb{Z})^{11}$ has order 2048. For six anchors $S=\{(0,0),(8,0),(16,0),(24,0),(32,0),(40,0)\}$, the action is **free**; the six orbits are pairwise disjoint and cover G (a 6×2048 tiling).

11.2 C768 schedule & closed forms

Implement the **canonical fair schedule** $\sigma_{\mathrm{CFS}}(t)=(t \bmod 48,\ t \bmod 256)$ of length 768. For the constant observable $f\equiv 1$, the closures are

$$S^{(0)}=768,\qquad S^{(1)}=294{,}528,\qquad S^{(2)}=150{,}700{,}160.$$

These serve as immediate regression tests.

11.3 Φ-compatibility (bulk–boundary)

Provide matrices M_p, M_b (over $\mathbb{F}_2$) and an anchor embedding ι so that Φ is **equivariant**: bulk toggles correspond to boundary XORs, and Φ-transported anchor orbits **partition** G .

11.4 R96 resonance classes

Run deterministic orbit reduction (fixed generator order + salt, Gray traversal). The normalized canonical representatives yield exactly **96 classes** with multiplicities $\{2{:}32,\ 3{:}64\}$. Normalization commutes with U_{ref} .

11.5 Audit bundle (pass/fail)

Deliver: config manifest, schedule digest, subgroup certificate, Φ-logs, and R96 enumeration. The verifier script must pass end-to-end.

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12) Proof Sketches & Guarantees

- **Contractivity ⇒ stability.** Under ACE with $\sum_p b_p|w_{p,t}|\le 1-\varepsilon_t$, we have $\|K_t\|\le 1-\varepsilon_t$ and $\rho(K_t)\le\|K_t\|_2<1$, giving **GapLB>0** and contraction on each audited step. WAC/DSE lifts this to trajectory convergence.
- **SPDE residuals.** Stationary OU forcing with Neumann boundary yields bounded expected residual energy under calibrated D ; audit compares empirical variance to SPDE forecast.
- **Subgroup tiling.** Independence of 11 commuting involutions gives order 2048; freeness on anchors implies six disjoint orbits covering G .
- **Φ-equivariance.** The defined linear XOR action commutes with Φ on anchors and extends by functoriality; hence Φ-transported anchor orbits partition G .

- **R96 determinism.** Fixed traversal + salt canonicalizes orbits; multiplicity pattern $\{2{:}32, 3{:}64\}$ follows from the normalization invariants and subgroup symmetry.

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13) Implementation Recipe (checkable)

1. **Enable UOR@IFMD:** adopt A9 + lattice; enforce the Processability Guard; provision QuantumFoam() + extractor; implement QCheck; define SlopeUB/GapLB and SPDE echo model; wire audit triggers and responses; deploy the monitor (ACE projection + auto-tighten).
2. **Bridge to runtime:** implement the Π-kernel adapter exposing $T_{t+1} = F_t + K_t T_t$, channelization, ACE projection to $\sum b_p |w_p| \le 1 - \varepsilon_t$, PETC logging, and WAC/DSE checks.
3. **Satisfy Φ-Atlas gates:** prove/order the 2048 subgroup + 6-orbit tiling; implement σ_{CFS} and verify closures; demonstrate Φ-equivariance; canonicalize to 96 resonance classes; ship the audit bundle.

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14) Reference Algorithms (pseudocode)

Weighted-ℓ_1 projection. Project w to $\sum_p b_p |w_p| \le \tau$ via a generalized soft-threshold with weight-dependent θ (sort by $|w_i|/b_i$). Enforce feasibility within numerical tolerance.

Spectral-norm SlopeUB and conservative GapLB. Estimate $\|K\|_2$ by power iteration on $K^\top K$; set $\mathrm{GapLB} = \max\{0, 1 - \|K\|_2\}$.

Audit loop. If SlopeUB $\ge$ threshold or GapLB $\le$ threshold, multiply τ by $\alpha < 1$, re-project, and re-estimate; log a minimally reproducible trace.

Adapter (projection-first loop). - Proposals $\widehat{w}_t$ from the kernel → governor scaling → ACE projection (budget $R_t = 1 - \varepsilon_t - X_t$). - Assemble $K_t = \sum_p w_{p,t} B_{p,t}$ and update $T_{t+1} = F_t + K_t T_t$. - Emit PETC entry; run WAC/DSE window checks.

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15) Notation Summary

ℓ_p, t_p, E_p, c : Planck units & light speed. L : lattice. Q_{K_δ} : measurement kernel with resolution δ . ACE: weighted-ℓ_1 projector and budget τ . SlopeUB: Lipschitz envelope. GapLB: $1 - \rho(K)$. PETC: prime-exact tensor consistency. WAC/DSE: window envelopes. U_{ref} : order-2048 boundary subgroup. σ_{CFS} : canonical fair schedule. Φ: bulk–boundary map. R96: resonance classes.

End of Note (APEX v1).